Cymbal Meet — Troubleshooting Guide: Device Performance

Diagnosing and Resolving Conference Room Device Issues

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1. Overview

Cymbal Meet conference room devices ("boxes") are purpose-built hardware units installed in meeting rooms. They provide high-definition video, spatial audio, and intelligent camera framing for in-room participants. When devices underperform, meeting quality degrades for all participants — both in-room and remote.

This guide covers the most common device performance issues, their root causes, diagnostic procedures, and resolution steps.

2. Performance Baselines

Understanding normal operating parameters is essential for identifying issues. Devices that consistently exceed these thresholds require investigation.

Normal Operating Parameters

Metric	Healthy Range	Warning Threshold	Critical Threshold
CPU Usage	20-45%	60-75%	>75%
Memory Usage	30-55%	65-80%	>80%
Network Latency	10-40ms	50-80ms	>80ms
Packet Loss	0.0-0.5%	0.5-2.0%	>2.0%
Video Quality Score	3.8-5.0	3.0-3.8	<3.0

Device Telemetry

Cymbal Meet devices report telemetry every 15 minutes during business hours (8 AM - 6 PM local time). Telemetry data includes CPU, memory, network, and video quality metrics. This data is available via the Cymbal Meet Admin Console and the BigQuery analytics pipeline.

3. Common Issues and Solutions

3.1 High Network Latency

Symptoms:

- Video freezing or stuttering
- Audio delay / echo
- Participants reporting "you're breaking up"

Video quality score consistently below 3.5

Causes (in order of likelihood):

Insufficient QoS (Quality of Service) Configuration

- The most common cause of latency issues, especially after office moves or network changes
- Video traffic competes with bulk data transfers, cloud backups, and software updates
- Impact: Latency spikes during business hours when network is congested

Network Congestion

- Too many devices sharing insufficient bandwidth
- Common when multiple conference rooms are in simultaneous use

Routing Issues

- Traffic routing through unnecessary hops (VPN, proxy, or remote gateway)

3.2 H[redacted]et Loss

Symp[redacted]

- A[redacted]g out (words/phrases missing)
- V[redacted]cts (blocky, pixelated video)
- S[redacted]ing showing stale content
- V[redacted]y score drops below 3.0

Root [redacted]

1. **N[redacted]frastructure Degradation**

   [redacted] switches or routers dropping packets
   [redacted]aded network buffers
   [redacted]x mismatch on switch ports

2. **B[redacted] Saturation**

   [redacted]raffic exceeds available bandwidth
   [redacted]on during peak hours or when large file transfers coincide with meetings

3. **V[redacted]terference** (if using WiFi backhaul)

   [redacted]eting WiFi networks on overlapping channels
   [redacted]wave ovens, Bluetooth devices, or other 2.4GHz interference

4. **I[redacted] Issues**

   [redacted]am packet loss beyond the organization's network boundary
   [redacted]g issues between ISP and Cymbal Meet's media servers

Diagn[redacted]s:

1. [redacted]ket loss trends in Admin Console > Devices > select device > Telemetry
2. If pac[redacted]
   - I[redacted]
   - C[redacted]
   - [redacted]
3. If pack[redacted]
   - Is[redacted]
   - C[redacted]
   - V[redacted]
4. If pack[redacted]
   - Li[redacted]
   - Ic[redacted]

Resolution [redacted]

1. **Immed[redacted]** Audio/[redacted] loss.
2. **Infrast[redacted]** bandw[redacted]
3. **Monito[redacted]**

3.3 Low V[redacted] Quality Score

Symptoms[redacted]

- Blurry [redacted]w-resolution video
- Came[redacted] switching to HD despite sufficient bandwidth
- Qualit[redacted]re consistently below 3.5 even with good network metrics

Root Caus[redacted]

1. **Bandw[redacted] allocation below HD threshold**
    - G[redacted]al Meet requires 2.5 Mbps per device for 1080p video
    - I[redacted]ated bandwidth drops below 1.5 Mbps, quality is capped at 720p
    - B[redacted] 0.8 Mbps, quality drops to 480p
2. **CPU t[redacted]ling**
    - D[redacted] CPU overheating, causing clock speed reduction
    - R[redacted] outdated firmware that has known performance regressions
3. **Camera ha[redacted]**
    - Lens o[redacted]
    - Came[redacted]
4. **Lighting co[redacted]**
    - Low ar[redacted]
    - Strong[redacted]

Resolution Step[redacted]

1. Verify band[redacted]
2. Check firmw[redacted] 4)
3. Check CPU [redacted] ventilation
4. Clean came[redacted]
5. If quality sc[redacted] diagnostic

3.4 High CPU [redacted]

Symptoms:

- Device UI is [redacted]
- Slow to join [redacted]
- Features lik[redacted]
- Device rebo[redacted]

Root Causes:

1. **Outdated fi[redacted]**
2. **Too many b[redacted]** sensors run[redacted]
3. **Hardware a[redacted]**

4. **L[redacted]ting rooms** — devices in room[redacted]n 20+ participants use more CPU [redacted]multi-stream p[redacted]

Resol[redacted]s:

1. U[redacted]ware to the latest stable relea[redacted]ee Section 4)
2. D[redacted]-essential features: Settings > [redacted]ures > disable Digital Signage, Room Sensors if not n[redacted]
3. Rebo[redacted] device (Settings > System > Restart) — resolves temporary memory leak[redacted]
4. If CP[redacted] upgra[redacted]

4. Firm[redacted]

Current [redacted]

Devic[redacted]
Cymbal [redacted]
Cymbal [redacted]
Cymbal [redacted]

Update [redacted]

1. Sche[redacted]
2. Admi[redacted]
3. Choo[redacted] hours[redacted]
4. Verif[redacted]

Known F[redacted]

- **v4.1.[redacted]** opera[redacted]
- **v4.0.[redacted]** telem[redacted]

5. Netw[redacted]

DSCP Ma[redacted]

Tra[redacted]
Audio ([redacted]
Video ([redacted]
Screen Sh[redacted]
Signaling [redacted]

Minimum [redacted]

L1 Support	Single-device issues, known issue workarounds		